Verbalized Probability

Verbalized Probability[58]

- **Pipeline:**
  - **Fine-tuning**

    Step 1: Sub-task Correctness Estimation

    Task: add/sub × 100(questions)

Prompt + $q_T \times 100$ → LLM → $a_M \times 100$

$$\hat{p}_T = \mathbb{E}_{q \in T}[\mathbb{I}(a_M)]$$

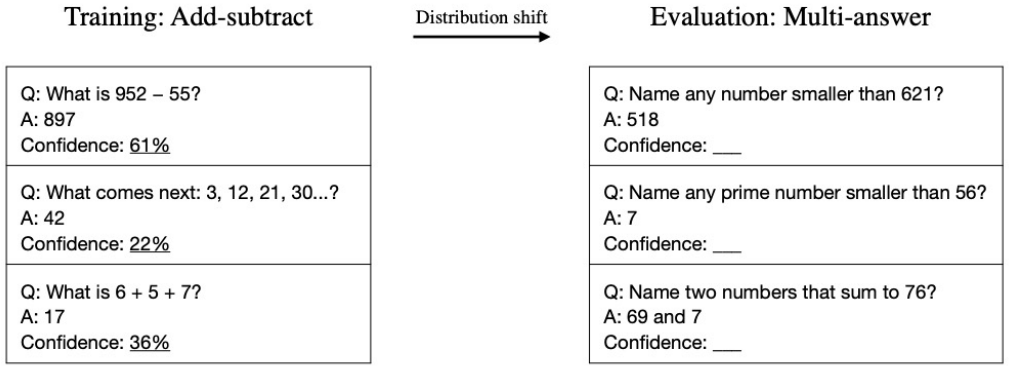
Step 2: Supervised Fine-tuning (SFT)

Input: Prompt+(q_T, a_M) Output: $\hat{p}_T$

  - **Generation (Inference)**

    Task: multi

Prompt + q_T → LLM(SFT) → A + P(Confidence)



Kind of probability	Definition	Example	Supervised objective	Desirable properties
Verbalized (number / word)	Express uncertainty in language ('61%' or 'medium confidence')	**Q: What is 952 – 55?** **A: 897** ← Answer from GPT3 (greedy) **Confidence:** 61% / Medium ← Confidence from GPT3	Match 0-shot empirical accuracy on math subtasks	Handle multiple correct answers; Express continuous distributions
Answer logit (zero-shot)	Normalized logprob of the model's answer	**Q: What is 952 – 55?** **A: 897** ← Normalized logprob for GPT3's answer	None	Requires no training
Indirect logit	Logprob of 'True' token when appended to model's answer	**Q: What is 952 – 55?** **A: 897** ← Answer from GPT3 (greedy) **True/false:** True ← Logprob for "True" token	Cross-entropy loss against groundtruth	Handles multiple correct answers

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- **Task Type:** Math generation
- **Introduction:** Verbalized Probability finetunes language models to output calibrated confidence statements by supervising regression targets derived from task-level empirical accuracy estimated via repeated zero-shot answers. It quantifies prediction uncertainty, providing a calibration mechanism independent of token logits within language-model uncertainty research.

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Single-pass	-	White-box	Fine-tuning	High	Calibration

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